Jianwei Jia

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EDUCATION

Georgia Institute of Technology, US | ECE (Ph.D.) **08/2023 - present**
GPA: 4.0/4.0 Supervisor: Prof. Shimeng Yu
The University of Michigan -Ann Arbor, US | VLSI (M.S.E.) **09/2021 - 04/2023**
GPA: 3.95/4.0
Nankai University, CN | Microelectronics (B.S.) **09/2017 - 06/2021**
GPA: 3.73/4.0

WORK EXPERIENCE

From 05/2026 to 11/2026 Circuit Design Research, TSMC, San Jose ***Research Intern***

RESEARCH & ACADEMIC EXPERIENCE

• Have Participated in Multiple Mixed-Signal IC Design and Digital IC Design Projects

1. Mixed-signal Circuit, Tape-out, and Test

From 08/2023 to 08/2025 Multiple FeFET Tape-outs, Gatech ***Research Assistant***
Led 3 successful design and tape-out chips, including the following projects: 1. A 28nm FeFET Analog CIM with On-Chip 4-bit Flash ADC; 2. aF-level Cap Measurement Circuit with Mixer; 3. FeFET TCAM with Low-power Technique; 4. Comparator Offset Cancellation Design based on FeFET. (PDK: GF28SLPe)

From 02/2024 to 05/2024 10-bit SAR ADC Tape-out, Gatech ***Team Leader***
Designed a 10-bit SAR ADC, including the bootstrapped circuit, 2-stage amplifier, strong-arm comparator, and a digital control block. (PDK: Ti lbc7, Sponsor: Ti)

From 01/2022 to 08/2022 Secure and Programmable PMU Tape-out, Umich ***Research Assistant***
Designed a Power Management Unit based on the switched-capacitor DC-DC converter generator under the open-source flow from OpenFASOC and OpenRoad. (PDK: skywater130, Sponsor: Google)

2. Digital Circuit Design

From 10/2022 to 12/2022 N-Way Super-Scalar with Early Branch Resolution, Umich ***Team Leader***
Designed an N-way super-scalar with early branch resolution under R10K architecture and RISC-V ISA.

From 10/2021 to 12/2021 Customized Floating-Point Unit, Umich ***Team Member***
Designed a 16-bit floating-point unit to realize addition, subtraction, and multiplication. (PDK: IBM130)

TECHNICAL SKILLS

Circuit Design Tools: Virtuoso, VCS, Design Compiler, Genus, Innovus, Prime Time, Altium Designer, etc.

Code Class Tools: System Verilog, Tcl, Spice, Python, Matlab, Makefile, C++, Linux, Perl, etc.

PUBLICATIONS

1. M. Chen, **J. Jia**, V. Garg, S. Yu, "**Ferroelectric Memories: Recent Chip-Scale Advances on Foundry Platforms**," in IEEE Solid-State Circuits Magazine (***SSCM***), 2026. (Accepted)
2. Y. Kong, H. Liu, **J. Jia**, S. Yu, "**RaceSAT: a 28nm Mixed-Signal Time-Domain k-Sat Solver with Inherent-Variation-Based Stochastic Annealing and ADC-Free Clause Evaluation**," in 2026 IEEE Asian Solid-State Circuits Conference (***ASSCC***), Taichung, Taiwan, 2026. (Accepted)
3. **J. Jia**, J. Sonawane, V. Garg, M. Chen, T. Xie, O. Phadke, Y. Kong, S. Li, S. Yu, "**aF-Resolution On-Chip Characterization of Sub-fF Non-Volatile Capacitance in Nanoscale FeFET**," in 2026 IEEE European Solid-State Electronics Research Conference (***ESSERC***), Palma de Mallorca, Spain, 2026. (Accepted)
4. P. Megginson, M. Vadlamani, **J. Jia**, G. W. Chapman, S. G. Cardwell, S. Yu, "**Charge-Domain Leaky-Integrate-and-Fire Neuron with Tunable Parameters Using Ferroelectric Non-Volatile**

Capacitors," in 2026 ACM International Conference on Neuromorphic Systems (***ICONS***), 2026. (Accepted)

5. W. H. Huang*, **J. Jia***, Y. Kong, F. Waqar, T. H. Wen, M. F. Chang, S. Yu, "**Hardware Acceleration of Kolmogorov-Arnold Network (KAN) in Large-Scale Systems**," in IEEE Transactions on Very Large Scale Integration (VLSI) Systems (***TVLSI***), April 2026.
6. M. Vadlamani, D. Chakraborty, **J. Jia**, H. Mulaosmanovic, S. Duenkel, S. Beyer, S. Datta, S. Yu, "**Cryogenic Characterization of Ferroelectric Nonvolatile Capacitors**," in IEEE Transactions on Electron Devices (***TED***), April 2026.
7. M. Chen, V. Garg, **J. Jia**, J. Sonawane, O. Phadke, S. Yu, "**Non-Volatile Digital Compute-in-Memory Macro with Ferroelectric FET-based Voltage Divider Weight Cells Featuring Power-Gating**," in IEEE Open Journal of the Solid-State Circuits Society (***OJ-SSCS***), March 2026.
8. V. Garg, **J. Jia**, O. Phadke, and S. Yu, "**A 28-nm FeFET Compute-in-Memory Macro With 64×64 Array Size and On-Chip 4-Bit Flash ADC**," in IEEE Solid-State Circuits Letters (***SSCL***), vol. 9, pp. 13-16, 2026.
9. **J. Jia**, Z. Jia, O. Phadke, Y. Shi, S. Yu, "**Reconfigurable Ferroelectric Bandpass Filter With Low-Frequency Noise Analysis for Intracardiac Electrogram Monitoring**," in IEEE Journal on Exploratory Solid-State Computational Devices and Circuits (***JXCDC***), vol. 11, pp. 67-73, 2025.
10. M. Vadlamani, **J. Jia**, T. Xie, Y.C. Luo, J. Lee, S. Li, S. Yu, "**Capacitive Crossbar Array for Solving Matrix Equations in One-Shot**," in IEEE Electron Device Letters (***EDL***), vol. 46, no. 3, pp. 389-392, March 2025.
11. Y. Kong, **J. Jia**, A. Lu, F. Waqar, Y.C. Luo, H. Li, I. Young, S. Yu, "**Digital Compute-in-Memory Ising Annealer with Ferroelectric Capacitor-Based nvSRAM for Combinatorial Optimization Problems**," in 2025 IEEE International Symposium on Circuits and Systems (***ISCAS***), London, UK, 2025.
12. W. H. Huang*, **J. Jia***, Y. Kong, F. Waqar, T. H. Wen, M. F. Chang, S. Yu, "**Hardware Acceleration of Kolmogorov-Arnold Network (KAN) for Lightweight Edge Inference**," in Proceedings of the 30th Asia and South Pacific Design Automation Conference (***ASPDAC***), New York, NY, USA, 2025.
13. **J. Jia**, Z. Jia, O. Phadke, G. Choe, Y. Shi, S. Yu, "**A Reconfigurable Bandpass Filter with Ferroelectric Devices for Intracardiac Electrograms Monitoring**," in 2024 IEEE 67th International Midwest Symposium on Circuits and Systems (***MWSCAS***), Springfield, MA, USA, 2024.